UPTIME ROBOTICS

BLOCK A CHALLENGE · BUAS × UPTIME ROBOTICS

The Weed Robot

Can a camera reliably spot a young weed in a paving joint?



2 November 2026 · Peter Brouwers & Jeroen Griffioen · Academy of AI, Games and Media

The next thirty minutes

01

Who we are

02

The problem

03

Our presumption

04

Your challenge

FIRST, A DISCLAIMER

So… not this one.

You are in the Netherlands, the name is ambiguous, and somebody was going to ask.



Two founders

Peter Brouwers

Markets and networks

Jeroen Griffioen

Strategy and ventures

UPTIME ROBOTICS

What we do

Keep robots running

Put robots to work

Today’s challenge sits in the second half.



PART ONE

A weed is a small problem. Ten million of them are not.

30 MARCH 2016

The chemical option disappeared

Everything since then only removes what is above the ground.



Nederlandse gemeenten geven bakken geld uit aan onkruidbestrijding

NIEUWS

Guus van Rijswijck, dinsdag 3 oktober 2017 1 sec

Sinds het wettelijk verbod op gif bij de bestrijding van onkruid zijn Nederlandse gemeenten bakken met geld kwijt aan het verwijderen van groen in de openbare ruimte. Gemeenten zijn miljoenen euro's kwijt; milieuvriendelijke alternatieven zijn nauwelijks effectief.

Stad+Groen, 3 October 2017

What the ban costs the Netherlands

€50–130m

extra every year, for municipalities alone. Our estimate, and we will show you how we got there.

2.5×

the cost per pass

€2.9m

a year in Rotterdam, up from 1.3

3 → 13

rounds a year in Hellevoetsluis

Municipal figures reported by Stad+Groen, 3 October 2017. The national range is our own estimate.



Photo: Omroep Brabant, 22 May 2026

And there is nobody left to do it

The method that replaced the sprayer needs hands.

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ProRail, Onkruidbestrijding

It is not only pavements

3,000 km of track

7,000 km of inspection path

2028 chemicals gone

Everyone is fighting grown weeds

Today

Heavy work on a plant that is already rooted.

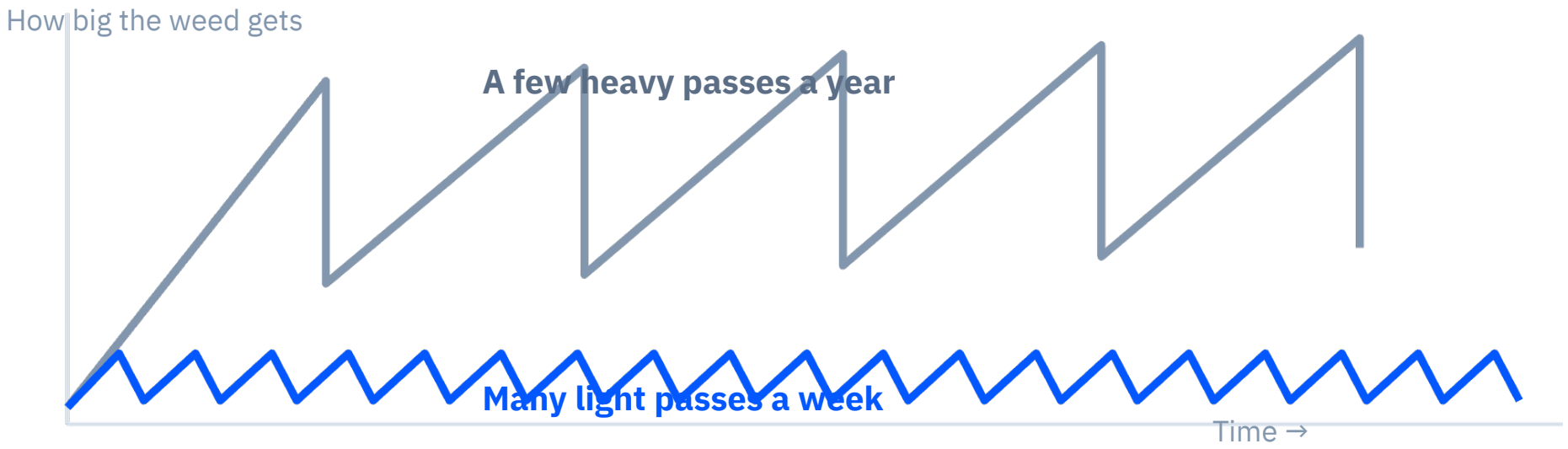
What nobody tries

A brush of contact on a seedling two days old.

PART TWO

Pass lightly, pass often, never let it grow

Change the interval, not the treatment



Four jobs. All four open.



See

Tell a weed from moss, gravel and a crack, on any paving and in any light.



Decide

Remember where weeds keep returning, and go back there more often.



Move

Cover a square or a car park alone, safely, around people and traffic.



Remove

End a seedling without damaging the joint or the paving around it.

We have opinions about all four. We are keeping them to ourselves until you have had your own ideas.

PART THREE

Everything depends on whether it can see

Detect weeds in paving, and tell us what it means

01

Make the dataset

No ready dataset exists. Collect, label and document your own.

02

Train the model

Classify, detect or segment. Your architecture, your call.

03

Measure it honestly

Precision and recall per paving type, plus a real error analysis.

04

Give us a verdict

Would you send this down a street unsupervised? Yes or no, and why.

Seventy per cent with an honest error analysis beats ninety-five on pictures we cannot trust.

The conditions are the problem

The surface	Clinkers, tiles, cobbles, gravel
The light	Sun, shadow, rain, dusk
The look-alikes	Moss, leaves, stains, planted green
The size	A seedling is a few pixels
The cost of a mistake	A miss and a false alarm cost different things

Being honest about where we are

No robot

No prototype, no customers, no orders. We have a problem we understand and a hunch about the answer.

Two founders on call

Mail us during the challenge and you get a real answer, not a brochure. Including the answers we do not like.

A real decision

In December we decide whether to build this at all. What you find goes into that decision.

If your conclusion is that a camera cannot do this reliably enough, that is a finding, and we want to hear it.

Your questions

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